OAKAI

OAKAI Development Workspace Architecture

Hermes Agent Dashboard UI — Holistic Architecture, Flow & Function

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Executive Summary

Hermes Agent workspace architecture — high-level view

The OAKAI development workspace runs on Hermes Agent — an autonomous AI agent operating inside a Docker-on-WSL2 container on the founder's Windows laptop. The workspace is organized into five architectural layers, each with distinct responsibilities. The agent is currently in the **foundation phase**: SSM registration pending, AI infrastructure fully operational on free-tier resources.

Key Metrics

Free-tier priority: Nous Portal ~50RPM, OpenRouter 50/day, NVIDIA NIM (one-time credit)

State protection: state.db gitignored + pre-commit hook blocks commits (prevents Aug 12 race condition)

Gateway service: s6-supervised, PID 46427, auto-restart enabled

Cron automation: 6 jobs, all model-pinned to prevent drift

This document provides both a holistic overview and in-depth technical details for each architectural layer, data flow, and operational function.

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Holistic Architecture Overview

Five-layer model with cross-cutting concerns

The workspace architecture is organized into five layers, from user-facing interface down to low-level persistence. Each layer encapsulates specific responsibilities and communicates with adjacent layers through well-defined interfaces.

(A)

Layer Summary

Layer	Component	Primary Function	Key Files
1. User Interface	TUI Dashboard, CLI, Web UI	Human interaction surface	tui-theme-boot.json, config.yaml
2. Agent Core	Hermes Agent, Skills Engine, Model Routing	AI reasoning + tool orchestration	config.yaml (agent.*), mem/, skills/
3. Tools & Gateway	Web, Browser, Vision, File, Message Gateway	External API + service integration	gateway_state.json, bin/tirith, .local/
4. Data & Knowledge	Knowledge Base, Cron Outputs, State DB	Information storage + retrieval	knowledge/, cron/output/, state.db
5. Persistence & Control	Git, Cron Scheduler, S6 Supervision	Version control + automation	cron/jobs.json, .git/, state.snapshots/

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(B)

Block Flow Diagram

(C)

Data Flow Diagram

Data Flow — User Request to Persistence

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User Interface Layer

How humans interact with the agent

(A)

TUI Dashboard

The Terminal UI (TUI) is the primary human-facing interface. It renders the agent's conversation, tool calls, and status in a rich color scheme defined by **tui-theme-boot.json**.

Property	Value	Purpose
Background	#1e1e1e (dark)	Eye comfort in low-light
Primary color	#FFD700 (gold)	Tool status / highlights
Accent color	#FFBF00 (amber)	Active selections / prompts
Border	#CD7F32 (bronze)	Panel boundaries
Text	#FFF8DC (cream)	High-contrast readable text
Status good	#8FBC8F (green)	Healthy / success state
Status bad	#FF6B6B (red)	Error / critical state

(B)

CLI & Web UI

CLI commands: **/help**, **/new**, **/stop**, **/status**. Web UI is provided via the browser-use cloud provider when GUI interaction is needed (e.g., form submissions, visual verification).

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Agent Core Layer

AI reasoning engine

(A)

Hermes Agent

The core agent runs with **max_turns: 150** per session. It loads skills from **/opt/data/skills/** and uses the model routing stack to select free-tier models first.

(B)

Model Routing (Free-First Priority)

Priority	Provider	Model	Quota	Notes
1st choice	Nous Portal	tencent/hy3:free	~50RPM	Default + cron-pinned
2nd choice	OpenRouter	poolside/laguna-s-2.1:free	50/day	Vision fallback
3rd choice	NVIDIA NIM	nvidia/nemotron-nano-12b-v2-vl	One-time credit	Vision only (-vl)
Paid (approval)	Direct	gemini-2.5-flash, deepseek-v4	Requires approval	Last resort

(C)

Skills Engine

Skills are Python scripts under **/opt/data/skills/**, organized by category: PDF generation, productivity (xlsx, powerpoint, docx), creative (design, diagrams), mlops (model routing, eval), research (web, arxiv), software development, and hermes-specific tooling.

Skill Category	Key Skills	Use Case
PDF	OAKAI Professional Engine, mission executive	Professional reports
Productivity	xlsx, powerpoint, docx, notion, obsidian	Daily deliverables
MLOps	free-tier-model-routing, hermes-cron-model-pinning	Cost control + automation
Creative	architecture-diagram, claude-design	Visual content

Skill Category	Key Skills	Use Case
Research	web-research-knowledgebase, arxiv, blogwatcher	Market intelligence

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Tools & Gateway Layer

External integration and messaging

(A)

Message Gateway (s6-supervised)

Property	Value	Function
Process	hermes gateway run --replace	Main gateway loop
Supervisor	s6	Auto-restart on crash
Gateway PID	46427	Currently running
Supervisor PID	46425	Health watchdog
Log PID	46428	Output capture
State	running	Active agents: 0 (standby)
Platforms	{} (empty)	No external platform connected yet

(B)

External Tools

Tool	Provider	Provider Tier	Status
Web Search	Firecrawl	Nous subscription	Active via gateway
Browser Automation	Browser-Use	Nous subscription	Cloud provider
Image Generation	FAL (FLUX.2)	Nous subscription	Active
TTS	OpenAI (edge/openai)	Nous subscription	Active via gateway
STT	OpenAI Whisper	Nous subscription	Active
Vision	NVIDIA Nemotron	NVIDIA (free)	Active (nemotron-nano-12b-v2-vl)
Local LLM	Qwen2.5-1.5B + bge-m3	Local Ollama	Running (egress fallback)

(C)

Tirith Threat Intelligence

The **tirith** binary (22MB) at **bin/tirith** provides threat intelligence. Its database lives at **.local/share/tirith/tirith-threatdb.dat**. It is referenced in config via **tirith_enabled** flag.

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Data & Knowledge Layer

How information is organized

(A)

Knowledge Base Structure

The knowledge base at **/opt/data/knowledge/** is indexed by **INDEX.md**. It follows a 4-bucket structure:

Bucket	Path	Purpose	Constraints
By Industry	knowledge/by_industry/	Client-specific POCs + domain research	≤32KB SUMMARY (hard cap)
Mentor	knowledge/mentor/	Daily AI education for founder	≤32KB SUMMARY (hard cap)
Raw	knowledge/raw/	Full fetched articles (traceability)	Not auto-loaded
Strategy	knowledge/strategy/	COO strategic guidance	Weekly generation

(B)

Cron Output Storage

Cron job outputs are stored at **/opt/data/cron/output//** as timestamped markdown files. Six active jobs:

Job ID	Schedule (UTC)	Schedule (MYT)	Deliverable
mentor-ai-daily	07:00, 11:00, 15:00, 19:00	15:00, 19:00, 23:00, 03:00	AI concept + test
learn-pensolar	07:00	15:00	PENSOLAR research update
workspace-cleanup-daily	02:00	10:00	14-day retention pruning
strategic-coo-guidance	Sun 00:00	Sun 08:00	Weekly COO brief
marketing-advisor-daily	06:00 (Mon-Sat)	14:00 (Mon-Sat)	Marketing content
startup-catchup-enforcement	06:00	14:00	Startup checks + catchup

(C)

State Management

File	Size	Purpose	Protection
state.db	~3.1MB	Session state, context	gitignored + pre-commit hook
kanban.db	14KB	Task tracking	gitignored
projects.db	4KB	Project registry	gitignored
verification_evidence.db	32KB	Test results log	gitignored
context_length_cache.yaml	227B	Token budget tracking	gitignored
.hermes_history	~varies	Chat session history	gitignored

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Persistence & Control Layer

Durability and automation

(A)

Git Version Control

Repository: **https://github.com/seowengguan-sudo/hermes-model-routing.git**. All changes are pushed ad-hoc (multiple commits). No daily sync cron job exists — pushes are performed manually after each significant change.

- Commit eb8adf2 — gitignore + pre-commit hardening
- Commit 2511c44 — PDF enhance with task deliverables
- Commit 5c299b5 — Mission executive PDF generator
- Commit 0fb524e — Deep research methodology skill
- Commit 3ff19f2 — Unified professional-doc-generation skill
- Commit b625a30 — Revised mission PDF with larger fonts
- Commit f75fa5a — Idempotent style registration fix
- Commit 1516e3f6 — INDEX.md cross-references update
- Commit 2e63059 — EG SEOW preferences + Gantt + architecture diagram
- Commit ae19735 — Aug 2026 calendar date/day correlation fix
- Commit 8b13142 — SKILL.md documentation update
- Commit d490ef2 — Layout optimization helpers + page-fit improvements

(B)

State Protection (Post-Aug 12 Incident)

August 12 Incident

Git auto-sync race condition zeroed state.db during concurrent read/write

Recovery: restored from commit 54c4651 (pre-race snapshot)

Fix: state.db added to .gitignore + pre-commit hook blocks commits

state.db is now untracked — protected from git operations

(C)

S6 Process Supervision

The gateway service runs under **s6** supervision with auto-restart. If the gateway crashes, s6 restarts it within 5 seconds. The supervisor and log processes ensure no silent failures.

Process	PID	Role
hermes gateway	46427	Main gateway loop (message delivery)
s6-supervisor	46425	Health watchdog + auto-restart
s6-log	46428	Output capture + logging

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Governance & Controls

Safety layers that prevent errors

(A)

Three-Layer Approval Framework

Gate	When	Mechanism	Status
Approval Gate	Before paid API calls	model-selection-policy skill	■ Active
Eval Gate	Before trusting AI output	golden_v1.csv + score_eval.py	■ Active
Autonomy Gate	Before AI acts autonomously	Shadow→Canary→Enforce rollout	■ Active

(B)

Pre-commit Hook

Located at **/opt/data/.git/hooks/pre-commit**. Blocks commits of:

- state.db and all SQLite database files (*.db)
- All credential/token files (.env, auth.json, *.txt secrets)
- Runtime cache files (context_length_cache.yaml)

(C)

Pre-commit Verification Results

Check Category	Checks	Status
State DB protection	Blocks state.db*, kanban.db*, projects.db*	■ 19/19 passed
Secret scanning	Blocks .env, auth.json, GithubToken.txt	■ Clean
Credential detection	Blocks API keys, tokens, passwords	■ Clean

(D)

.gitignore Coverage

Protected Pattern	Examples
Runtime databases	state.db*, kanban.db*, projects.db*, cron/executions.db*
Logs	logs/*, *.log
Credentials	.env, GithubToken.txt, auth.json, *.credentials
Caches	context_length_cache.yaml, .hermes_history
Session state	desktop/interrupted_turns.json, skills/.usage.json

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Deployment & Infrastructure

Runtime environment details

(A)

Environment

Property	Value
Host	Windows (WSL2)
Container	Docker (daemon currently down — irrelevant)
Python	3.13.5 (no pip module, PEP 668)
Package Manager	uv (installed)
Working Dir	/opt/data
Git Remote	github.com/seowengguan-sudo/hermes-model-routing
Profile	default (local)

(B)

Directory Structure

Directory	Purpose	~Size
.venv (architecture/hermes-venv)	Python env with reportlab, openpyxl, pymupdf	86MB
lazy-packages/	Isolated package dir (pymupdf, reportlab, pyyaml)	1.1MB
knowledge/	Knowledge base (INDEX.md + 4 buckets)	varies
skills/	Agent capabilities (categorías, hermes, productivity...)	100+ skills
cron/	Scheduled automation (jobs.json + output/)	1KB + outputs
workspace/	Generated deliverables (PDFs, data)	varies
state-snapshots/	Periodic recovery snapshots	post-Aug12 recovery

(C)

Egress Constraints

Network Reality

Egress IP: 161.142.137.99 (TTNET, Penang MY)

Cloudflare WAF blocks Groq/Cerebras (HTTP 403/1010)

HuggingFace DNS-blocked (NXDOMAIN for *.huggingface.co)

Nous API rate-limited (but still accessible — primary provider)

Free-tier priority: Nous → OpenRouter → NVIDIA NIM → Paid (approval-gated)

Mitigation: local-first stack (Qwen2.5-1.5B + bge-m3 via Ollama)